

THE PENNY MAGAZINE



Issue 32

Monthly Supplement for September, 1832

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1 TO SEPTEMBER 29, 1832 ***

Monthly Supplement of

THE PENNY MAGAZINE

OF THE

Society for the Diffusion of Useful Knowledge.

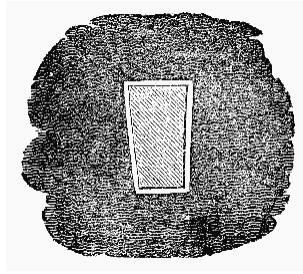
32.]

SEPTEMBER 1 TO SEPTEMBER 29, 1832

THE THAMES TUNNEL.



[Transverse Section of the Thames Tunnel.]



[Section of Drift-way.]

As far back as the year 1802, a project was set on foot by some enterprising gentlemen, with a view of opening an archway under the Thames, between Rotherhithe and Limehouse, not far from the line of the present tunnel. The engineer selected for this enterprise was particularly qualified for such an undertaking, being an experienced Cornish miner. Having made some borings at the Horse-ferry and on the

opposite side of the river, he reported that "he was firmly persuaded the undertaking would not cost so much as had been conceived." A subscription was, in consequence, raised; and a company was formed, under the denomination of the "Thames Archway Company." Surveys, plans, and estimates were made, and an Act of Parliament being obtained, the work was begun. The engineer commenced operations by sinking a shaft of 11 feet diameter, at 330 feet from the line of the wharf on the Rotherhithe side. But the obstacles which he encountered from the nature of the ground increased to such a degree, as he proceeded, that at the depth of 42 feet he was obliged to desist. A subsequent report of borings, however, having proved very favourable, an enterprising proprietor engaged to complete the shaft (reduced to 8 feet diameter) to 76 feet, at which depth it was discovered that it would be dangerous to go deeper. At this stage of the proceedings, viz. in August, 1807, a second engineer was engaged by the Company, a gentleman whose name had been coupled with very great enterprises in the mining department. Before opening the drift-way both engineers agreed to reduce its breadth to 2 feet 6 inches at the top, and 3 feet at the bottom. At the depth of 76 feet they found the ground to consist of a *firm dry sand*; and there they opened the drift, which they carried forward in a gentle ascent. In November, 1807, when 394 feet of the drift had been completed, the services of the first engineer were dispensed with after four years and a half of hard labour. The Directors then agreed to give the second engineer £1000, by way of premium, if he succeeded in reaching the opposite shore. The drift was further extended to 814 feet, through equally firm dry ground, with the precaution, which had been employed

from the beginning, of a substantial planking all the way. One hundred and thirty-eight feet more were cut through a bed of calcareous rock eight feet thick. But on the 21st of December, the head of the drift had hardly entered two feet into the stratum, which lay immediately over the rock, when the roof broke down in a loose state, leaving above head a cavity large enough for a man to stand in it. It is to be observed that there was no less than thirty feet of intervening ground between the drift and the river at the time this accident happened. The engineer succeeded in filling and securing the cavity; but, such was the nature of the whole ground above the rock, that, under the influence of an extraordinary high tide (on the 26th of January 1808), the ground again made its way fast in a loose state into the drift, and the river soon broke through 25 feet of ground. This same tide caused the destruction of the Deptford and Lewisham bridges. The engineer having succeeded in filling and closing this hole, the miners re-entered the drift, which was reduced to three feet in height, for the purpose of clearing the dangerous place. The miners had, therefore, to work on their knees: however, notwithstanding every effort to attain the opposite shore, they were driven away by the frequent bursts of sand and water. The engineer having afterwards sounded the ground from above, reported that he had no doubt the two fractures communicated underneath; and therefore admitted that it was quite impracticable to go further except by means of a cofferdam or caissons. On the 30th of March, 1809, the Directors offered a reward for the most approved plan of completing the archway. Fifty-four plans having been obtained by this announcement, they were referred to the opinion of scientific men. These gentlemen reported that they were unanimously of opinion, that an archway, of any useful size, was impracticable under the Thames by an underground excavation on any of the plans that had come before them; observing, at the same time, that they did not pretend to assign limits to the ingenuity of other men. A further trial was made by a third engineer, who operated from above the river, but it proved equally fruitless. Thus ended in 1809 all the exertions and the efforts made during nearly seven years, for the purpose of accomplishing an archway under the bed of the Thames; at the end of which period not so much as a drain had been completed, nor had the miners succeeded in working in any of those strata wherein the excavation for the archway must eventually have been effected.

Several years afterwards, Mr. Brunel was prevailed upon by one of the most active promoters of the archway enterprise (Mr. J. Wyatt) to turn his

attention to the subject; and, being furnished with the documents connected with the first attempt, he devised his plan with the impression that both the excavation and the structure might be made on a full scale at once.

Before proceeding to an exposition of the plan adopted by Mr. Brunel, and of the means by which he has carried it into execution, we have to state that the structure of the Thames Tunnel, as represented in the annexed view, is 38 feet in width and 22 feet 6 inches in height externally and that a length of 600 feet, in the style of a double arcade, has been made, though one arch only is open to public inspection. The excavation therefore made under the Thames for this structure presents a sectional surface of 850 feet, which is equal to sixty times the area of the drift; but for a more comprehensive demonstration of this aggregate, the reader is to understand that the excavation which has been made for the Thames Tunnel is larger than the House of Commons, the dimensions of which are 32 feet in width and 25 feet in height, presenting therefore only 800 feet of sectional surface. At high water, the head of the river is about 75 feet above the foot of the excavation, and consequently three times the height of that room. These circumstances, independently of the nature of the ground, are sufficient to place the work of the Thames Tunnel among the boldest enterprises in the art of engineering.

Notwithstanding that the first attempt had contributed to discourage all idea of success, there were still sufficient evidences to indicate that by beginning in the stratum of dry firm sand, and keeping close under the stratum of clay forming the bottom of the river, there was space enough to effect the object, although the nature of the intervening ground had been ascertained to be very loose in many places. All the information obtained from the miner's report concurred with the opinions of geologists in pointing out that the most eligible line for the Tunnel was to keep as near to the bottom of the river as the security of the work would permit. The first idea of the plan which appeared to the engineer best calculated for making an excavation fit for the object under so overwhelming a head of water, was suggested by the sight of a piece of a keel of a ship which had been eroded by the operation of the worm called the terido. From this he conceived it practicable, as his specification describes it, to make a circular opening of sufficient capacity at once. However, of the two modes which he described, he gave the preference to that of proceeding by forming simultaneously several contiguous excavations by means of an apparatus which has been

denominated the shield. This shield upon the whole partakes of the character of a powerful cofferdam, applied in a horizontal instead of the vertical direction. It consists of twelve parallel frames lying close to each other like so many volumes in a bookcase. Each frame, being nearly 22 feet in height, is divided into three stories: the whole presents therefore thirty-six openings or cells. It is from these cells that the miners, operating by small quantities at a time, like so many teridos, are able to erode the ground in front, while others at the back bring up a substantial incrustation, namely the brick structure. For locomotive action each frame is provided with two substantial legs resting on equally substantial shoes, (not unlike snow-shoes); these legs are provided with articulations that fit the frames for a pacing movement. The shield has perambulated 600 feet of its assigned career; and has left behind a substantial structure in the form of a double arcade, which is now as long or nearly as the Burlington Arcade.

With regard to the external form of the structure, and the mode adopted for its execution, it must be obvious to persons acquainted with such matters, that the most substantial form, and the best calculated at the same time to prevent, as far as practicable, any derangement in alluvial strata of various degrees of density, is the square form, as corresponding with that mode of building which is technically called underjoining and underlaying. Thus, in fact, the bed of the river, with its contents, has been underlayed, just as the Customhouse has been, to receive the superstructure.

An indispensable requisite in a work of this nature was, that it should be made proof against the greatest disasters that were to be apprehended, notwithstanding every precaution that could be taken. Mr. Brunel's plan was considered by his Grace the Duke of Wellington, by Dr. Wollaston, and by those engineers and scientific men who had the opportunity of examining the designs, and of hearing the description given by the engineer, as being well calculated to accomplish the contemplated object, although some apprehensions were raised at the time as to what might result from so formidable an occurrence as an irruption of the river, considering the extent of the devastation it might cause in the ground and among the works. The engineer afforded such explanations as allayed, in some degree, those apprehensions which, it must be admitted, he has since completely dispelled by undeniable facts.

It was under these auspices that the plan was brought before the public in 1823, and that in the month of February, 1824, subscriptions were obtained to a large amount to carry it into effect, notwithstanding the novelty of the scheme, and its risks.

[To be continued.]

BRITISH BIRDS.

[An Outline of the Smaller British Birds, by Robert A. Slaney, Esq. M.P.]

This little volume is evidently the production of an intelligent, humane, and truly amiable mind, whose enthusiasm on a favourite study is calculated to win attention and to enamour others of the same subject.

It is painful to reflect, that while we are everywhere surrounded by natural objects, animate and inanimate, each replete with interest and beauty of its own, so many of us should be ignorant, and thence insensible and heedless of these charms. This negligence is almost culpable, for so slight is the expense of time and study required to render ourselves familiar with many of the works of nature, that the busiest and poorest could afford the sacrifice, and thus put themselves in possession of a pure and cheap source of enjoyment. But old writers encumbered the simplicity of natural history with Latin and Greek terms, and words of *interminable* length, making appear difficult and abstruse what in reality is not so. It has been the laudable object of Mr. Slaney, as of other writers of the present day, to make these matters easy, as they should be. In the little volume now before us, our readers will find every thing perfectly intelligible to the commonest capacity, and its attentive perusal will delight and enrich their minds with numerous interesting accounts of the formation and habits of the beautiful winged creatures we every day see flitting around us, while, by indicating the course and modes of observation to be pursued, it will facilitate their acquirement of other and more extensive branches of natural history.

In allusion to the general ignorance on these subjects even among the wealthy and educated classes of society, Mr. Slaney says very amusingly, “We have sometimes asked our fair young friends if they knew as many of the smaller birds as they could count on their fingers? They usually answered confidently in the affirmative, but could seldom get much beyond one hand.”

Now this ignorance, which is mere negligence, we again say is almost culpable, and, as it tends to remove this ignorance by a single and delightful process, we therefore warmly recommend this ‘Outline of the Smaller British Birds.’ The reader will find in it, beside Mr. Slaney’s own descriptions and observations, many amusing and instructive extracts from some of the most distinguished writers on natural history, such as White of Selborne, who has so admirably described what passed under his own eyes in a province of England; Wilson, originally a common weaver of Paisley near Glasgow, who found an unknown world of beauty in the birds of the American wilds, his accounts of which unite the accuracy of philosophy with the beauty and glow of poetry; Knapp, the author of the ‘Journal of a Naturalist;’ the late Sir Humphry Davy, &c. &c.

The following extracts will give an idea of the work:—

“Perhaps, if we take a short view of our common birds, beginning with the missel thrush, the largest British songster and coming down to the golden-crested wren, it may repay our trouble. Within these limits we shall find about seventy birds, varying in size, form, habits, structure, and notes: most of which are seen, at one or other time of the year, in the fields and woods which surround our dwellings, and many of them are constantly with us. They may be divided into hard-billed birds, feeding on grain, seeds, and fruits; and soft-billed birds, on insects and worms. Some feed on both; and many grain-eaters devour insects, though few of the soft-billed eat seeds. These are again divided into families, from some peculiarity in their formation (chiefly the beak): as the finches, buntings, warblers, &c.

“In considering the birds to which we have confined our view, we find they divide themselves into three sets—winter visitors, summer visitors, and sojourners.

“The smaller winter visitors, about five in number, come to our hospitable shores in autumn, and leave us in the spring. They all come from colder climates; and as the frost locks up their sources of subsistence in the north, (where, in summer, they have built their nests and reared their young,) led by that wonderful instinct which their Maker has implanted, they direct their airy flight across the mountain and the flood.

“The summer visitors, on the other hand, coming to us in the spring, and leaving us in the autumn, all come from the south; and depart again to the regions of the sun as winter approaches.

“The winter visitors are all (except the grey wag-tail), hard-billed birds, fitted to feed on seeds, berries, and fruits found during our winters. They are chiefly gregarious, and seem by their numbers to band themselves together against the piercing season!”

The Fieldfare, the Redwing, the Starling, are the commonest of these winter visitors.

“Let us turn to our summer visitors. They come to us in the spring, as the weather becomes warmer, the earth clothed with vegetation, and the air and surface of the ground begin to teem with insect life; when the chrysalis bursts its case, the worm, and slug, and caterpillar, ‘and every creeping thing after his kind,’ come forth; then appear, led by an unseen hand, myriads of soft-billed warblers from distant lands, formed to thin the insect race, and whose services warmly deserve our gratitude and protection.

“From March till May ten thousand busy pinions ply the air, by day and night, and bring those melodious visitors from all the southern countries, where the parching heat at this season renders their food difficult to procure. As they arrive, they disperse throughout the country,—

‘They to their grassy couch, these to their nests.’

Each grove and shrubbery, each ‘bosky dell from side to side,’ each heath and upland common, each hedge and garden and petty rural homestead, receives some of these wandering minstrels. It is probable they return year after year, if undisturbed, to the same haunts; and perhaps revisit with as much pleasure as ourselves the well-known scenes of their youth—

‘When Nature pleased, for life itself was new,
And the heart pictured what the fancy drew.’

“We may smile at the idea of fancy or feeling in a bird; yet those who have closely watched those beautiful beings will readily believe as much difference in their dispositions as Cowper found in the temper of his hares. The ‘mellow lark who at Heaven’s gate sings’ must be endowed with instincts—superior to those of the ‘poor beetle that we tread upon.’ Memory birds possess in a considerable degree. Swallows will choose out the same nook for their nest year after year. That elegant little bird, the common fly-catcher, is attached to the same spot. A pair built for three summers successively in the same place, close to the writer’s study window; and their chase for gnats and other insects was under his view, as he sat reading; and for a considerable period the parent birds, ‘from early dawn till latest eve,’ might be observed catching assiduously our English muskitoes.”

Among the most interesting of these summer visitors are the nightingale, the whitethroat, the redstart, the fly-catcher, the swallow, and the cuckoo.

The third division, viz., sojourners or resident birds, are then considered. Of these only five belong to the soft-billed birds, namely, the robin and the wren, the hedge-sparrow, and the black and white water wagtail. In the classes of hard-billed birds always with us, are the nuthatch, the blackbird, the thrush, the greenfinch, the bullfinch, the goldfinch, the chaffinch, the sparrow, the lark, and others too numerous to mention here, but which will be found classified, and very ably described in Mr. Slaney’s volume.

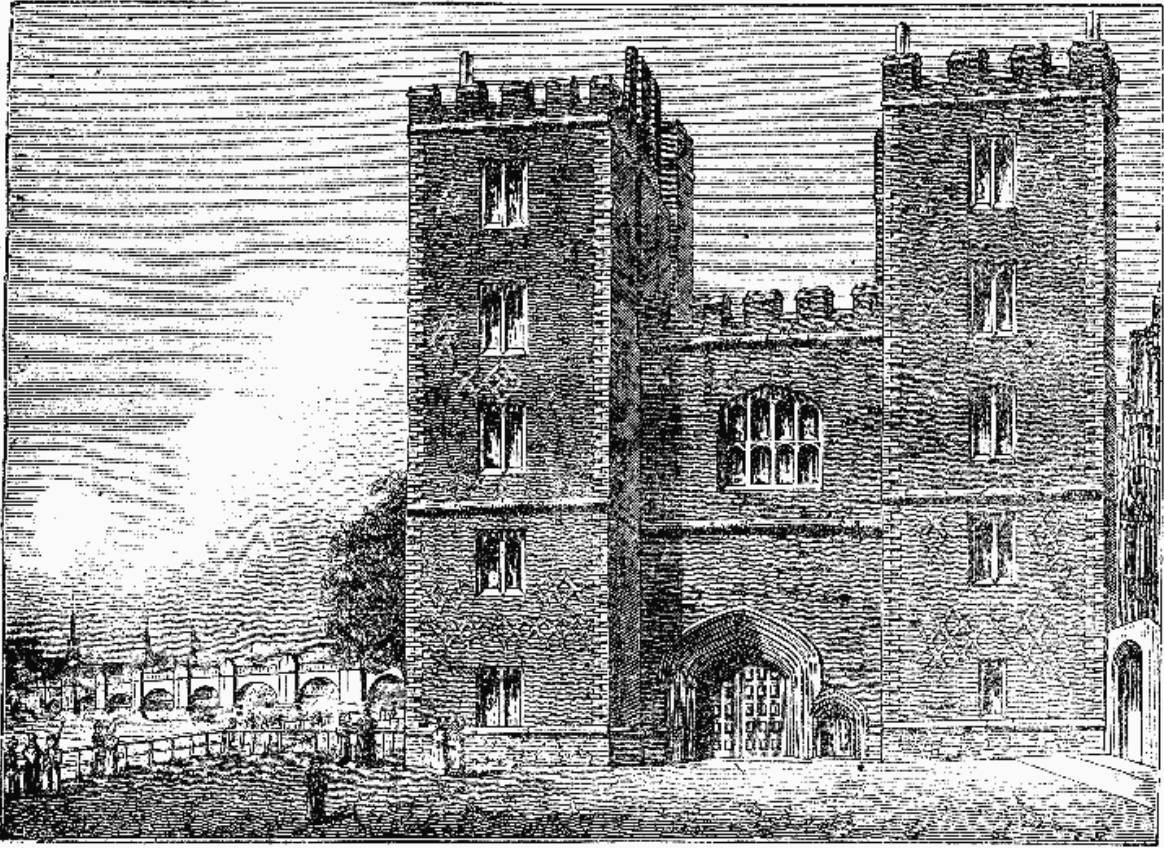
“About twenty song birds of passage come to us, and rear their young in our island. Of these some are local species, and some but partially and thinly scattered. These guests of summer remain to enjoy our finest weather, when the warmth of the climate, and the richness of vegetation, and the

harmony of nature, invite us abroad. We think that our fair readers might double the pleasure of their walks if they knew each note of their tiny visitant, and distinguished the form and plumage of every feathered songster.—p. 12.

“The nightingale is celebrated in all countries: its sober plumage of tawny brown would never attract our attention, though its light and elegant form might excite admiration. This delightful songster is not found north of Shrewsbury in the west, or Doncaster in the east; and is seldom seen in Devonshire or Cornwall. ‘It has been observed that it is not seen but where cowslips grow plentifully,’ indicating a damp, cool soil, and probably yielding those insects it delights in. All writers praise the song of this bird. We will only quote the eloquent expressions of a naturalist (Wilson) called forth by a songster of the new world:—

“‘When every object around conveys the sensation of joy, and heaven’s abundance is, as it were, showering around us the grateful heart beats in unison with the varying elevated strains of this bird. We listen to its notes in a kind of ecstasy as a hymn to the great and most adorable Creator of all. Abject must that heart be, and callous those feelings, and depraved that taste, which neither the charms of nature, nor the melody of innocence, nor the voice of gratitude or devotion, can reach.’”—
p 16.

LAMBETH PALACE.



[Gate-house of Lambeth Palace.]

Lambeth Palace, which stands on the right bank of the Thames, within half a mile of Westminster Bridge, has been for many centuries the principal residence of the Archbishops of Canterbury. The manor belonged originally to the see of Rochester, to which it had been granted, before the Norman Conquest, by a sister of Edward the Confessor; and it was obtained in exchange for some other lands, by Baldwin, Archbishop of Canterbury, in the year 1189. There is reason to believe, however, that the Archbishops had a house here for at least a century before this time. The ancient possession of Lambeth by the see of Rochester is still commemorated by the payment to the latter, in two half-yearly sums, of five marks of silver, in consideration of the lodging, fire-wood, forage, and other accommodations which the Bishops of Rochester had been accustomed to receive here whenever they visited London. When the Archbishops of Canterbury first obtained possession of the place, the buildings on it appear to have been old

and mean. With the exception of the Chapel, the whole of the present structure has certainly been erected since the middle of the thirteenth century.

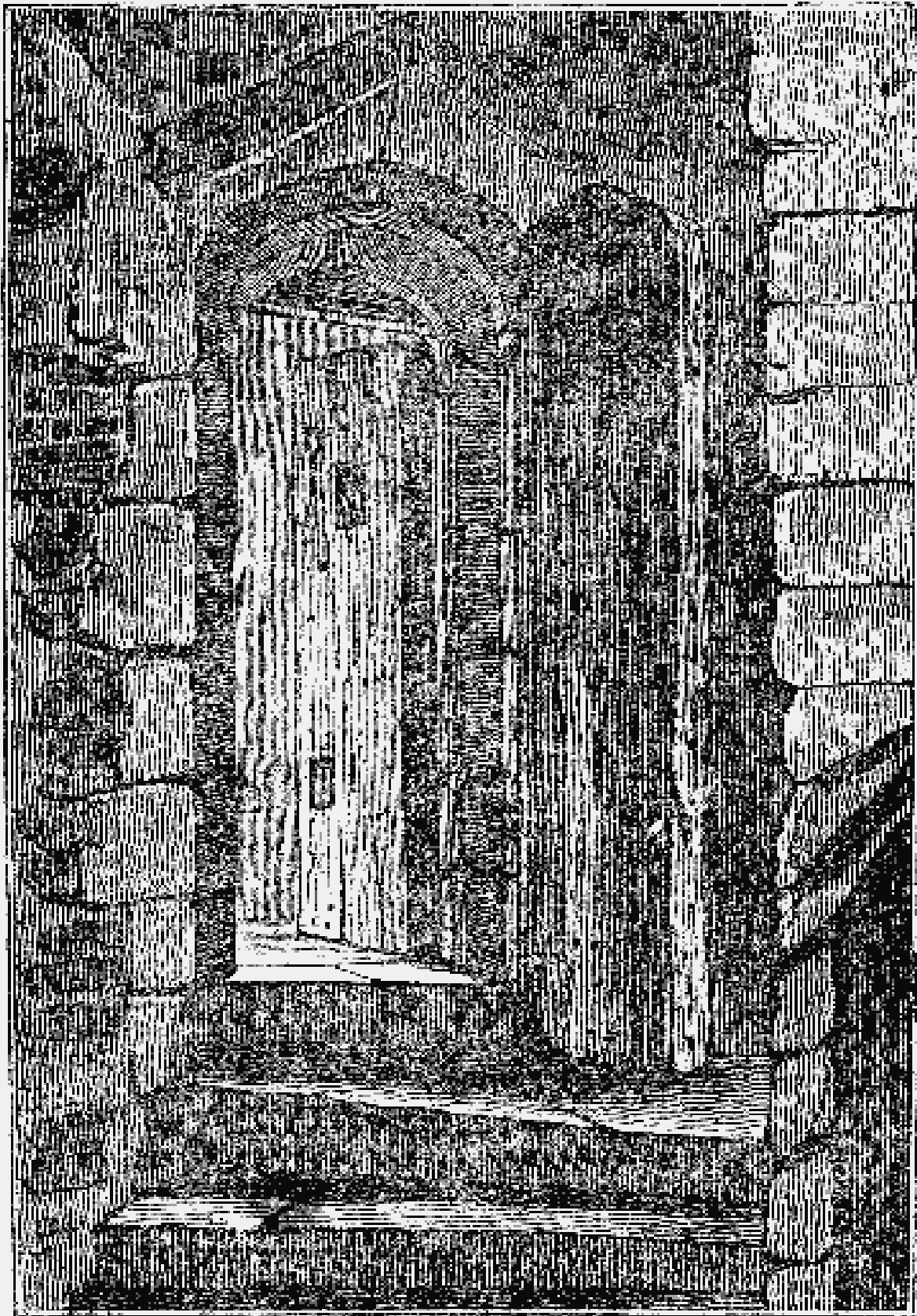
The Palace, as it now appears, is an irregular but very extensive pile, exhibiting specimens of almost every style of architecture that has prevailed during the last seven hundred years. The oldest part of it, as we have just said, is the Chapel, which is supposed to have been built towards the close of the twelfth century. It consists of two apartments, divided by a richly-ornamented screen, and measuring together 72 feet in length by 25 in breadth. The height of the Chapel is 30 feet. Under it is another apartment of smaller dimensions, formed by a series of arches, supported by pillars, and now used as a cellar, though in ancient times it may not improbably have served as a place of worship. Another of the most remarkable portions of the edifice, the Great Hall, was originally erected by Archbishop Chicheley in the beginning of the reign of Henry VI.; but after the Palace had been sold by the Parliament in the time of the Commonwealth, this magnificent apartment was pulled down. It was rebuilt, however, on the old site, and in close imitation of the former hall, after the Restoration, by Archbishop Juxon, at an expense of £10,500. It stands on the right of the principal courtyard, and is built of fine red brick, the walls being supported by stone buttresses, and also coped with stone, and surmounted by large balls or orbs. The length of this noble room is 93 feet, its breadth 38, and its height 50. The roof, which is of oak and elaborately carved, is particularly splendid and imposing. The Gate-house, which forms the principal entry to the Palace, was erected by Cardinal Morton, about the year 1490, and is a very beautiful and magnificent structure. It consists of two lofty towers, from the summits of which is one of the finest views in the neighbourhood of the metropolis. In front of this gate the ancient archiepiscopal *dole*, or alms, is still distributed every Sunday, Tuesday, and Thursday, to thirty poor parishioners of Lambeth. Ten are served each day, among whom are divided three stone of beef, ten pitchers of broth, thickened with oatmeal, five quartern loaves, and twenty-pence in copper.

One of the most interesting portions of Lambeth Palace is the stone building called the Lollard's Tower. It was erected by Archbishop Chicheley, in the early part of the fifteenth century, as a place of confinement for the unhappy heretics from whom it derives its name. Under the tower is an apartment of somewhat singular appearance, called the *post*

room, from a large post in the middle of it by which its flat roof is partly supported. The prison in which the poor Lollards were confined is at the top of the tower, and is reached by a very narrow winding staircase. Its single doorway, which is so narrow as only to admit one person at a time, is strongly barricaded by both an outer and an inner door of oak, each three inches and a half thick, and thickly studded with iron. The dimensions of the apartment within are 12 feet in length, by 9 in width, and 8 in height; and it is lighted by two windows, which are only 28 inches high, by 14 inches wide on the inside, and about half as high and half as wide on the outside. Both the walls and roof of the chamber are lined with oaken planks an inch and a half thick; and eight large iron rings still remain fastened to the wood, the melancholy memorials of the barbarous tyranny whose victims formerly pined in this dismal prison-house. Many names, and fragments of sentences, are rudely cut out on various parts of the walls.

Among the other principal apartments are the Library, containing a very extensive and valuable collection of books and manuscripts, founded by Archbishop Bancroft in 1610: and the Long Gallery, generally supposed to have been the work of Cardinal Pole, who possessed the see from the death of Archbishop Cranmer in 1556 till 1558. This noble room contains many portraits, of which several are in the highest degree interesting as works of art, or on account of the individuals whom they represent.

Besides these apartments, the palace contains many others well deserving of notice, but which we cannot here attempt to describe. We may merely mention the Guardroom, an ancient and venerable chamber, 56 feet in length, and adorned by a splendid timber roof; the Presence Chamber, also of considerable antiquity: the great Dining-room, which contains a series of portraits of all the Archbishops, from Laud to Cornwallis inclusive; the old and new Drawing-rooms, the latter a fine room measuring 33 feet by 22, built by Archbishop Cornwallis; and the Steward's Parlour, probably built by Archbishop Cranmer. In a future number we shall describe the extensive additions to Lambeth Palace which have been made by his Grace the present Archbishop. The palace is surrounded by a park and gardens, very tastefully laid out, and occupying in all about eighteen acres. Among the ornaments of the grounds are particularly deserving of notice two Marseilles fig-trees, of great size, and still bearing an abundance of delicious fruit, which tradition asserts to have been planted by Cardinal Pole.



[Doorway in Lollard's Tower.]

EMIGRATION.

[Hints on Emigration to Upper Canada, &c., by Martin Doyle. Price One Shilling. Second Edition, 1832.]

The name of Martin Doyle is no doubt familiar to many of our readers, and if it is not, we hope to make it so by directing their attention to a sensible little book on Emigration to Upper Canada. It is chiefly addressed, as its title tells us, to “the middle and lower classes of Great Britain and Ireland,” who being blessed with health, strength, and the laudable desire of improving their condition, meditate a trip across the Atlantic. Whether the causes which annually induce so many thousands to quit their native shores are likely to increase or diminish, we can yet hardly form a judgment, but this we do know, that while the industrious emigrant, who is hard pressed at home, will undoubtedly be a gainer by the change, and while the province of Upper Canada is daily receiving a vigorous and healthy population into her bosom, we of the old country cannot at present consider ourselves otherwise than as losers by the change. It may be urged that a new market will be created in the Canadas for the products of British industry, and that thus emigration may ultimately, and indeed in no short period, prove beneficial to the mother-country. Such advantages we may reasonably expect, if the produce of Upper Canada is at all times freely allowed to enter our markets; but in the mean time we lose the honest and industrious labourer, we lose the small capitalist who has been struggling against difficulties at home, while the lazy, the vicious, and the good-for-nothings are left behind.

But as people *will* emigrate with the hope of improving their condition, it is not fair to grumble at them for trying to mend their fortune. Let us rather give them every reasonable facility for doing so, and above all let us

encourage every attempt to show them what is really the state of an emigrant to our American provinces, and what qualifications are essential to ensure his success. This our friend Martin, we think, has successfully accomplished in a little shilling book, in which he treats of the soil, climate, animal and vegetable productions, of Upper Canada, mingling with his remarks on these subjects sound sense and good advice. His picture of the fertility of these countries must make a hungry labourer's mouth water,—to think of the plenty of Indian corn, of wheat, oats, barley, &c., all raised on a virgin soil, with no labour but the sowing, after the necessary preliminaries of cutting down the trees or barking them are finished. We do not think that the products of Upper Canada are at all overrated by our author, save perhaps in the article of fruits, when he tells us (p. 42) that “pine-apples are raised without trouble, and melons and grapes grow wild in the woods:” it is of very little importance whether pine-apples can be easily raised or not, as we are persuaded turnips will prove a more profitable crop,—but we doubt the fact. As to the wild melons and the grapes, there is no danger of a man eating them in a country where bread, meat, and potatoes abound; otherwise we might put in a word of caution against them.

Though Martin Doyle has perhaps not been at all inclined to underrate the advantages open to an emigrant into Upper Canada, he has at the same time carefully avoided all such false views as might seduce the man who is comfortable at home to try the discomforts inseparable from settling in a new land. We fully agree with him in advising those who are well off at home to stay there, and to weigh all matters carefully before they think of leaving it. The industrious labourer who struggles hard against poverty, the skilful carpenter or shoe-maker, and others of the same class, may, by industry and perseverance, provide for their families in Canada, and find a refuge from the threatening evils of poverty. The single man who is disposed to the same adventure may lay the foundation of wealth and respectability by a few years of steady labour in the new world, But there is *one* condition on which we are glad to see that Martin Doyle has most strongly insisted—*sobriety*; which everywhere is an essential, but in North America it is all in all. In a country where the cheapness of ardent spirits, the very general want of some comfortable beverage such as beer, and the contagion of bad example, are all combined, it is indispensable that the emigrant arm himself with every possible precaution against the assaults of the beastly and degrading vice of drunkenness. Dram-drinking spreads

disease, poverty, vice, and misery, from the fertile banks of the North American lakes to the warm and tropical shores of the Mexican gulf: it is destroying more than war and famine cut off in older and less favoured parts of the world. Let then the drunkard stay at home to finish his short and filthy course: he will not improve his condition or his health by a settlement in the Canadas. We sincerely hope that the efforts of many excellent and enlightened men, both in North America and elsewhere, are even now doing something towards eradicating the vice of drunkenness.

We believe Martin Doyle's little book contains as much information as a man can require who thinks of going to the Canadas, and we strongly recommend its perusal to them as well as to others who are interested in the subject of emigration. As far as we are able to judge, the facts are in general correct, and have been collected with all proper diligence and discrimination. The advice is sound and wholesome; and a spirit of benevolence breathes through the whole. Not having space for any extracts, we must limit ourselves to noticing a mode of preparing milk for sea use, which is recommended to those who take provisions for the voyage. "Milk, after having been boiled should be carefully sealed up in jars, and if loaf sugar be added to it, there is no danger of its not keeping fresh during the voyage."

THE STEAM ENGINE.

[Lectures on the Steam-Engine, by the Rev. Dionysius Lardner, LL. D. 4th Edit. 12mo. London, 1832. Taylor. Price 7s. 6d.]

This is decidedly the best popular account which we have of the Steam-Engine,—the most accurate and complete and at the same time the most intelligible and interesting. The book is intended for the use of unscientific readers, or of the public generally; but it has the unusual advantage, for a work of this description, of being written by a person of the highest

scientific acquirements. The utmost dependence, therefore, may be placed on the correctness of the various descriptions and explanations it presents; they are stripped indeed of those technicalities which would have rendered them both repulsive and obscure to ordinary readers,—but the substance of the statement which remains, although thus transformed from a full and precise detail to a general outline, is still, in so far as it goes, perfectly clear and satisfactory. The volume contains all that most persons, with the exception of engineers and mathematicians, can be supposed desirous of knowing about the exceedingly curious and important subject of which it treats. It both describes the steam-engine as it now exists, and it relates its origin and past history. After a preliminary lecture on the natural forces on which the action of the contrivance depends, the author proceeds to recount and describe the successive inventions and improvements of the Marquis of Worcester—of Captain Savery—of Newcomen and Cawdley—of Humphrey Potter (the boy who first hit upon the plan of making the engine work its own valves),—and finally of the illustrious Watt. The various parts of Watt's single and double-acting engines are explained with the fulness which their importance deserves in the fifth and the three following lectures, the modifications which since his time have been introduced in the valves, the boiler, the furnace, &c., being also noticed. The ninth lecture is occupied with an account of the double-cylinder engines of Hornblower and Woolf, and also of a very ingenious and elegant machine, possessing certain peculiar and valuable recommendations, suggested by the late Reverend Mr. Cartwright, the well-known inventor of the power-loom. These expositions are in general distinguished by very remarkable clearness—and, illustrated as they are by abundance of excellent woodcuts and copper-plates, they can hardly fail to be perfectly and easily understood even by those who have been but little accustomed to such investigations.

The part of the book, however, on account of which we principally notice it at present, is the new matter which appears for the first time in this edition, and which relates to the two latest among the various extraordinary applications of the power of steam;—the wonders that have been performed by it on the Liverpool and Manchester railway, and the still more recent experimental achievements of the several projectors who have for some time past been endeavouring to introduce steam-coaches upon our common roads. The two long chapters which are devoted to these subjects, the first

especially, contain a number of facts which had not previously been laid before the public, and some of which are of considerable interest.

The locomotive (or travelling) engines which are employed on the Liverpool railway are all what are called high-pressure engines. One of Watt's most ingenious contrivances was his condensing apparatus, by which, previous to every stroke of the piston, he created a vacuum in the part of the cylinder through which it had to be driven, and thereby enabled it to be sent forward through that space with a much inferior pressure of steam to what would otherwise have been required. But in the steam-engines affixed to coaches it is found convenient to dispense with this apparatus, on account of its complexity, its weight, the room which it would occupy, and above all the constant supply of cold water which would be requisite to keep it in action. The consequence is that in these engines and others similarly constructed, a much greater force of steam is necessary to make the piston do its work; and they are on that account denominated high-pressure engines. It is only within the last thirty years that they have been introduced, and the most remarkable proofs of their power have been afforded on the Liverpool railway, which was opened only about two years ago.

Some time before it was opened a contest took place on this railway between three different steam-coaches. The Rocket, constructed by Mr. Stevenson; the Sans-pareil, by Mr. Hackworth; and the Novelty, by Messrs. Braithwait and Ericson, for a prize of £500, offered by the Directors to that which should accomplish the greatest distance in the shortest time. On this occasion the Rocket, which gained the prize, went over the space of 30 miles in 2 hours 6 minutes and 49 seconds, being at the average rate of $14\frac{1}{5}$ miles per hour. But in the course of the journey it sometimes proceeded at the rate of above 29 miles an hour, and its slowest motion was about $11\frac{1}{2}$ miles in that time^[1]. In May last, Dr. Lardner saw the engine called the Victory draw on the same rail-road the weight of 92 tons 19 cwt. 1 quarter, in twenty waggons, together with its tender containing fuel and water, from Liverpool to Manchester, a distance of 30 miles, in 1 hour 34 minutes and 45 seconds, besides 10 minutes spent in taking in water. The speed on this occasion was in some places $25\frac{1}{2}$ miles an hour, and on level ground, where there was no wind, it was generally 20 miles an hour. On another day, the engine called the Samson drew fifty waggons laden with merchandise, and, with itself, making a gross weight of above 233 tons, the same distance in 2

hours and 40 minutes, exclusive of delays upon the road for watering, &c., the rate of motion having varied from 9 to 16 miles an hour, and being on an average nearly 12 miles an hour. The coke consumed in this journey was 1762 lbs., or a quarter of a pound per ton per mile. The attendance required is only an engine-man and a fire-boy, the former of whom is paid 1s. 6d. for each trip, and the later 1s. The expense of the original construction of the engines, however, and of their wear and tear is very great, though not nearly so great on the latter account, Dr. Lardner assures us, as has been sometimes stated. The price of one of the most improved engines at first is about £800, and it will travel from 25,000 to 30,000 miles without costing as much more for repairs. Notwithstanding many extra expenses which this undertaking, as the first of the kind, has had to bear, and from the experience purchased by which future speculators will profit, it has been perfectly successful in a commercial point of view. The profits on the capital invested have been from the first above 6 per cent. per annum; and during the latter six months of 1831, it was at the rate of above 8 per cent. per annum; and it has since probably exceeded that amount. The original £100 shares already sell for above £200. On the other hand the advantages to the public have been as great as to the proprietors. Fully 600,000 passengers now pass by the rail-road in the course of the year between Liverpool and Manchester, or four times as many as used formerly to make the journey. The transference of merchandise is also effected both with infinitely greater speed, and at a vast reduction of expense.

Some time ago a work was published by Mr. Gordon, the engineer, on the application of steam as a moving power for coaches on common turnpike roads, the facts contained in which were principally derived from the report on this subject of the Committee of the House of Commons, which was ordered to be printed on the 12th of October last. Dr. Lardner has here availed himself of the information supplied by the same most interesting and important parliamentary paper; some of the curious details given in which we may possibly take another opportunity of laying before our readers. In the meanwhile we can only afford room for the general conclusions to which the Committee came on the evidence brought before them. They are as follows:—

“1. That carriages can be propelled by steam on common roads at an average rate of ten miles per hour.

- “2. That at this rate they have conveyed upwards of fourteen passengers.
- “3. That their weight, including engine, fuel, water, and attendants, may be under three tons.
- “4. That they can ascend and descend hills of considerable inclination with facility and safety.
- “5. That they are perfectly safe for passengers.
- “6. That they are not (or need not be, if properly constructed) nuisances to the public.
- “7. That they will become a speedier and cheaper mode of conveyance than carriages drawn by horses.
- “8. That, as they admit of greater breadth of tire than other carriages, and as the roads are not acted on so injuriously as by the feet of horses in common draught, such carriages will cause less wear of roads than coaches drawn by horses.
- “9. That rates of toll have been imposed on steam-carriages, which would prohibit their being used on several lines of road, were such charges permitted to remain unaltered.”

The toll-bills complained of have since been repealed. “At the moment that I write,” says Dr. Lardner, “several steam-carriages are in process of construction for regular work upon the public roads of England. Some are about to be established between Paddington and the Bank, upon the New Road; others between London and Greenwich, and other places in the vicinity of the metropolis. Another it is stated, is to run between London and Birmingham. The first impulse once received, the progress will be rapid, and the effect of proportionate importance.”

[1](#). In another place Dr. Lardner states that “the engine which conveyed Mr. Huskisson to Manchester, after the unhappy occurrence which took place at the great trial, moved at the rate of 35 miles an hour.” (p. 206.)

THE HALL OF ELTHAM PALACE.

Near the village of Eltham in Kent, about eight miles south east from London, stood anciently one of the most magnificent of the English royal palaces. The property is ascertained to have belonged to the Crown in the time of the Saxons. The Conqueror granted it to one of his Norman followers; but having again been forfeited to the Crown, it was given by

Edward I. to one of the most powerful barons of those times—John de Vesci. Soon after this it came into the possession of Anthony Bec, the famous military Bishop of Durham, who is accused, however, of having made the acquisition by the most shameless violation of his trust, as guardian of the legal heir. Bec is the earliest proprietor of the manor who is recorded to have erected any buildings on the site of the palace—although there can be little doubt that there was a house there before his time. He built a large and splendid mansion, which appears to have been completed soon after the middle of the thirteenth century, King Henry III., accompanied by his queen and all the principal nobility, having kept his Christmas here in 1269. This was probably the warming of the house. On the death of the Bishop, which took place here in 1310, the manor of Eltham fell again to the crown, in the possession of which it has ever since remained. For the next two centuries the place was a favourite residence of our monarchs. Edward II.'s son John was born here in 1315, and was thence called John of Eltham. In the reign of Edward III. the Parliament was on several occasions assembled at Eltham; and here that prince, in 1365, entertained his captive, John, King of France, with sumptuous hospitality. The palace was almost entirely rebuilt by Edward IV., who, on the conclusion of the work in 1482 is recorded to have kept his Christmas in the Hall with great state and splendour. Large additions were afterwards made to the building by Henry VII., who, like his predecessors, generally lived here, and was wont to dine every day in the hall, surrounded by his barons. At this time the royal palace of Eltham consisted of four quadrangles enclosed within a high wall, beyond which was a moat of great width: the whole formed an irregular area, approaching to a square in shape, to which the principal entry was over a bridge and through a gateway in the north wall. There was also another bridge and gate at the opposite side of the inclosure. Of the buildings the most important part consisted of a high range which crossed the court from east to west, and included the hall, the chapel, and the state apartments. To the palace were attached a garden and three parks, comprehending together above 1300 acres, besides the demesne lands of 400 acres more. These parks were stocked with deer, and many fine old trees that still remain testify how richly wooded the place must have formerly been.

Of all this magnificence but little now remains; and many parts of the buildings can scarcely be traced even in their foundations. Henry VIII.

deserted Eltham for the new palace of Greenwich, which, as being nearer to London, was probably found to be a more convenient residence. After this Eltham was only occasionally visited by the sovereign; which it sometimes was even in the time of James I. On the establishment of the Commonwealth the place was seized by the parliament and sold; and at the same time the parks were broken into, and the deer dispersed and killed by the soldiers and the common people. The work of devastation, thus begun, was continued until the greater part of the palace also was first reduced to a heap of ruins, and then swept away altogether; and although the property was recovered by the crown at the Restoration, no pains seem to have been taken to save the remnant of the pile from spoliation and destruction. On the contrary, the business of demolition was now carried on upon system: the old palace was turned into a quarry; and stone after stone was carried away as it was wanted for even the meanest purposes, until scarcely anything remained which it was thought worth while to remove. Fortunately, it was considered that the hall would make a good barn; and to this ignoble appropriation, which so well hit the economical humour of the times, we owe the preservation, in a state of comparative entireness, of this principal and most interesting portion of the noble old palace of Eltham.



[Hall of Eltham Palace.]

According to Mr. Buckler, who published a valuable account of this palace a few years ago, the length of the hall in the inside is above 101 feet by about 36½ in width. “The interior,” says this writer, “is magnificent. The taste and talent of ages are concentrated in its design; and it is scarcely possible to imagine proportions more just and noble, a plan more perfect, ornaments more appropriate and beautiful; in a word, a whole more harmonious than this regal banquetting-room.” The windows, which, however, have been long built up, are ranged in couples along both sides; and each series is terminated by a bay window at the west or upper end of the hall. But the most conspicuous ornament of this fine room is its splendid roof. “The main beams of the roof,” says Mr. Buckler, “are full 17 inches square and 28 feet long, perfectly straight and sound throughout, and are the produce of trees of the most stately growth. A forest must have yielded its choicest timber for the supply of this building; and it is evident that the material has been wrought with incredible labour and admirable skill.”

This hall was undoubtedly the erection of Edward IV., whose well-known symbol, the expanded rose, is still to be seen on various parts of the building. About four years ago the public attention was called to the state of this beautiful remnant of our ancient architecture, which it was understood there was an intention of levelling with the ground, on the pretext that the roof in some parts showed signs of decay, and threatened to fall if not taken down. It had been resolved, it seems, to remove the roof to some new building at Windsor or else where, and then to demolish the rest of the hall. By the exertions, however, of some individuals of taste and influence a reconsideration of the subject was obtained; and eventually it was determined by the Government to advance a small sum, in order to effect such a partial repair of the hall as might at least secure its stability for the present. The work was committed to the superintendence of Mr. Smirke, by whom it was executed with much ability; and the roof is now once more restored, as far as was practicable, to its original strength and beauty. In other respects, however, the hall, we believe, still remains in the state in which it was previous to the repairs and continues to be used as a barn.

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Transcriber's Notes

New original cover art included with this eBook is granted to the public domain. Itemized changes from the original text:

- [p. 257](#): Replaced “succeedeed” with “succeeded” in phrase “The engineer having succeeded in filling and closing this hole.”
- [p. 258](#): Replaced single with double opening quotation mark before phrase “The smaller winter visiters, about five in number.”
- [p. 259](#): Added closing quotation mark after phrase “catching assiduously our English muskitoes.”
- [p. 259](#): Added opening square bracket before subheading “Hints on Emigration to Upper Canada.”
- [p. 260](#): Replaced comma with period after phrase “the Archbishops had a house here for at least a century before this time.”
- [p. 260](#): Replaced “Chichely” with “Chicheley” in phrase “It was erected by Archbishop Chicheley, in the early part of the fifteenth century.”
- [p. 262](#): Replaced comma with closing parentheses in phrase “The locomotive (or travelling) engines which are employed on the Liverpool railway.”
- [p. 264](#): Added period after phrase “restored, as far as was practicable, to its original strength and beauty.”

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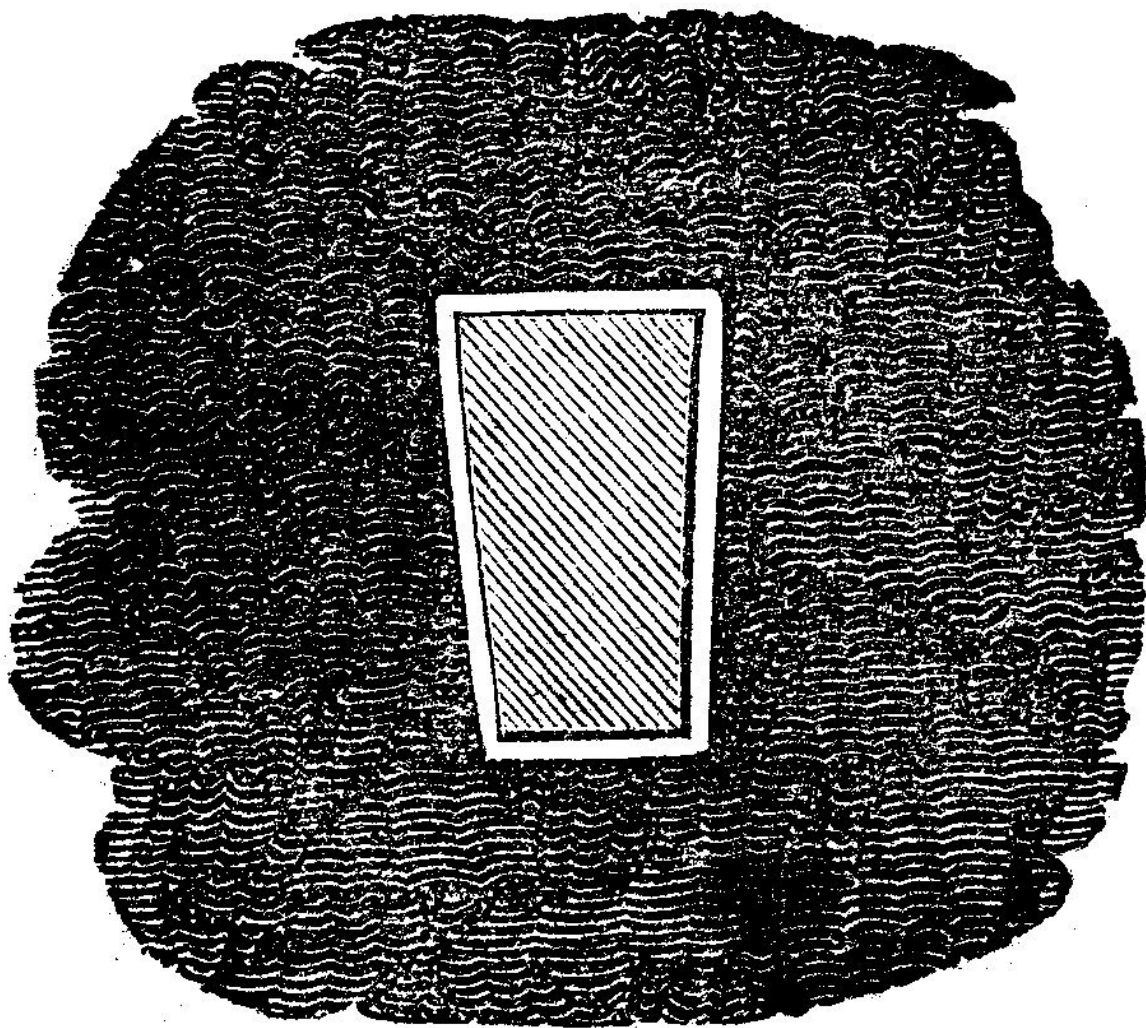
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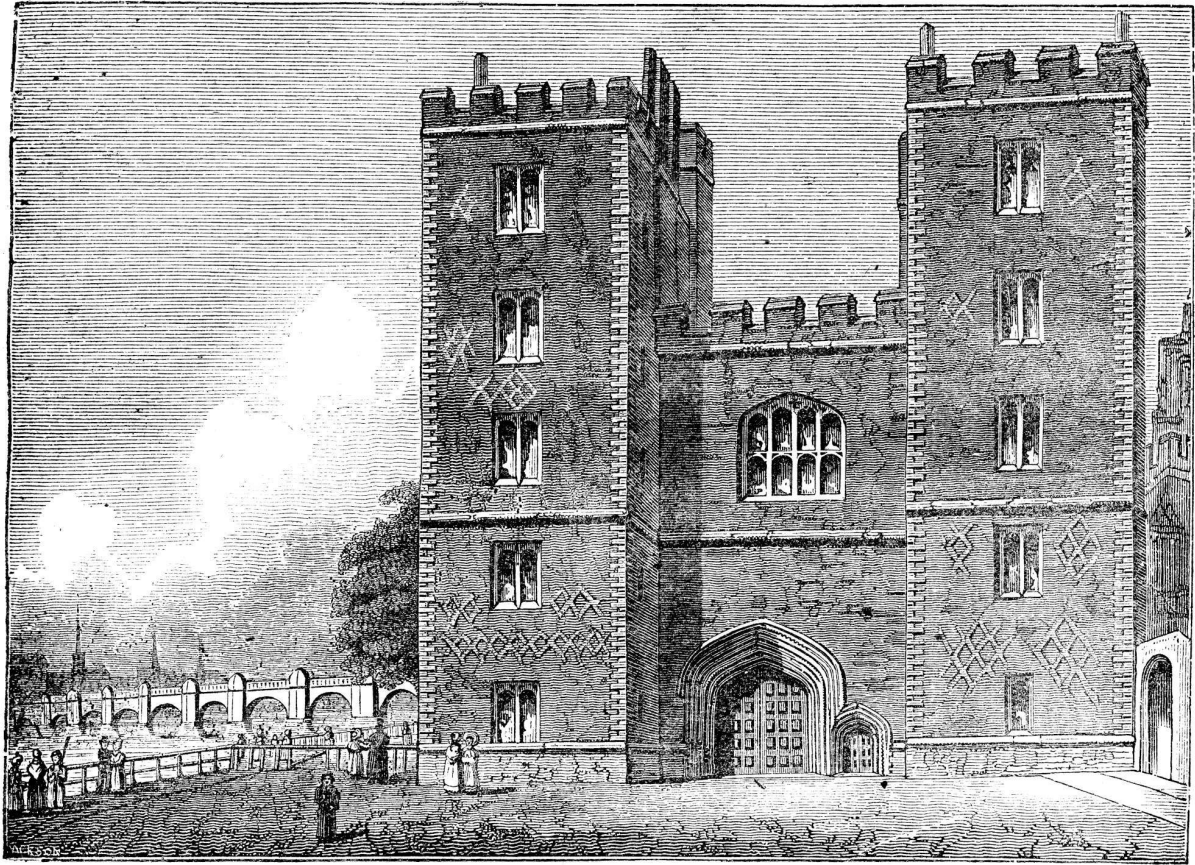
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